The Modality Gap in Visual Safety: A Usability-Aware Comparative Study of Local Multimodal LLMs

Junhua Tang
Department of Computer Science
University at Buffalo
junhuata@buffalo.edu

CSE 674: Final Report
Selected Project Option: 1. Comparative Analysis

Abstract

Multimodal large language models (MLLMs) inherit strong text-side alignment from modern LLMs, but recent work shows that this safety alignment does not reliably transfer to visual inputs. We present a usability-aware comparative study of three recent local MLLMs, Qwen3.5-9B, Gemma-3-12B, and GLM-4.6V-Flash, under three representative visual jailbreak attacks: FigStep, SceneTAP, and Beyond Visual Safety (BVS). Using MM-SafetyBench, we run a full baseline sweep across 13 risk scenarios with 10 samples per model-attack-scenario condition, yielding 117 harmful-evaluation conditions and 1,170 attack queries, and we complement this with a 390-query benign utility benchmark. To make this study practical, we implement a cascading evaluator that combines rule-based refusal detection with a Gemini-based LLM judge and spot-check auditing. Our results show that vulnerability is strongly attack- and scenario-dependent: Health Consultation emerges as the most consistently vulnerable scenario, SceneTAP and FigStep expose substantial jailbreak success on several conditions, and BVS is especially effective against GLM-4.6V-Flash while remaining much weaker on Qwen and Gemma. Benign utility results reveal that low attack success does not necessarily imply high usability cost, with Gemma achieving zero measured benign over-refusal on our control split. Finally, a focused CoCA follow-up on completed BVS-vulnerable conditions shows non-uniform defensive behavior: CoCA improves some conditions, leaves others unchanged, and backfires on several GLM cases. These findings suggest that multimodal safety evaluation should jointly measure vulnerability, refusal behavior, and defense side effects rather than relying on attack success rate alone.

1 Introduction

Multimodal large language models (MLLMs) are increasingly deployed in settings that require grounded reasoning over both text and images, including visual assistants, content moderation pipelines, and domain-specific copilots. Although recent language models have achieved strong safety alignment in text-only settings through instruction tuning and preference optimization, this protection does not automatically extend to multimodal inputs. When harmful intent is embedded in an image rather than written directly in text, an MLLM may treat that content as neutral visual evidence instead of as a dangerous instruction. This gap creates a meaningful attack surface for visual jailbreaks.

Recent multimodal safety research has shown that harmful instructions can be hidden in synthetic typographic images, blended into realistic scenes, or reconstructed through semantically indirect visual compositions. These attacks are concerning not only because they can increase compliance with unsafe requests, but also because they expose a measurement problem: a model that avoids direct compliance may still fail to issue an explicit refusal, while a more conservative model may over-refuse benign inputs and degrade usability. As a result, reporting attack success rate (ASR) alone provides an incomplete picture of real-world multimodal safety.

This project studies that gap through a systematic comparative benchmark of three recent local MLLMs: Qwen3.5-9B, Gemma-3-12B, and GLM-4.6V-Flash. We evaluate them under three representative visual jailbreak families, FigStep, SceneTAP, and Beyond Visual Safety (BVS), across all 13 MM-SafetyBench risk scenarios. We additionally measure benign over-refusal on a control evaluation split and run a focused follow-up with Constitutional Calibration (CoCA) on completed BVS-vulnerable conditions. The final benchmark covers 117 harmful conditions and 39 benign conditions, all evaluated with fixed seeds, identical prompting conventions, and a shared judge pipeline.

Our study is motivated by two practical gaps in prior work. First, many papers emphasize single-attack or single-model results, which makes it difficult to distinguish model-specific weakness from attack-specific artifacts. Second, large-scale multimodal safety evaluation is expensive when every model response must be judged by an external LLM, which discourages broader scenario coverage. To address these issues, we pair a broad empirical comparison with an efficient cascading evaluation design that uses a fast refusal-pattern filter before escalating ambiguous cases to a Gemini judge with spot-check auditing.

The main contributions of this report are as follows:

- We present a comparative multimodal safety benchmark spanning three local MLLMs, three visual jailbreak attacks, and 13 risk scenarios, with 1,170 harmful evaluation queries and 390 benign utility queries.
- We use a usability-aware evaluation protocol that jointly reports attack success rate (ASR), refusal rate (RR), and a Safety-Usability Index (SUI), supported by a cascading rule-plus-LLM judge pipeline.
- We provide a focused defense follow-up showing that CoCA is not uniformly beneficial in multimodal safety settings: it improves some BVS-vulnerable conditions, has no effect on others, and worsens several completed GLM cases.

Overall, our results argue for evaluating multimodal safety as a joint problem of attack robustness, refusal behavior, and benign utility. A model can look strong under one attack, fragile under another, and still exhibit a very different usability profile on benign prompts. This variability is exactly why localized, attack-specific claims should be treated cautiously when assessing the safety of modern MLLMs.

2 Related Work

Foundational multimodal architectures. The modern MLLM landscape was enabled by representation-learning and fusion advances such as CLIP [1], Flamingo [2], BLIP-2 [3], LLaVA [4], and InstructBLIP [5]. These systems show that frozen or partially frozen vision encoders can be effectively connected to powerful language backbones through contrastive pretraining, cross-attention, Q-former modules, or instruction tuning. At the same time, mainstream alignment methods such as RLHF and instruction-following optimization were developed primarily for text-based interactions [6]. This creates an important asymmetry: strong text-side helpfulness and harmlessness do not guarantee equivalent robustness once harmful intent is routed through visual tokens.

Visual jailbreak attacks. Recent work has made this asymmetry concrete through several families of multimodal jailbreaks. FigStep [7] uses typographic prompts to exploit the OCR ability of vision-language models and convert image text into actionable harmful instructions. SceneTAP [8] extends that idea by embedding adversarial typography into realistic scenes, making attacks less visually conspicuous and more plausible in real-world environments. BVS [9] further broadens the threat model by reconstructing harmful intent from semantically indirect visual cues rather than relying only on readable embedded text. Jailbreak in Pieces [10] similarly shows that compositional multimodal prompts can bypass safeguards that would likely hold in text-only settings. While these works establish the existence of visual jailbreak risk, they often emphasize the success of a single attack family or a limited model set. Our study instead compares multiple attack mechanisms under a unified experimental protocol.

Safety benchmarks and evaluation criteria. MM-SafetyBench [11] provides a structured benchmark for evaluating harmful multimodal behavior across diverse risk categories, while Omni-SafetyBench [12] extends safety evaluation to broader audio-visual settings. These benchmarks are valuable because they move evaluation beyond ad hoc jailbreak demonstrations. However, reporting ASR alone can still miss important alignment behavior. REVEAL [13] highlights the broader need to reason about both safety and utility, especially when refusal-heavy policies mask underlying model limitations. This perspective is relevant to our setting because multimodal models may respond with benign but noncommittal descriptions, explicit refusals,

or actionable compliance, and these behaviors should not be collapsed into a single success/failure view.

Inference-time defense and automated judging. Because retraining large multimodal models is costly, many recent defenses operate at inference time. CoCA [14] calibrates model behavior with constitutional prompts to recover safety awareness, while methods such as Immune [15] and ASTRA [16] attempt to suppress harmful continuations through alignment-aware generation control. Yet the effectiveness of these methods across different multimodal architectures and attack types is still underexplored. In parallel, automated evaluation itself has become a practical challenge: LLM-based judges improve nuance and coverage, but they also introduce latency and API cost. Our work contributes to this methodological space through a cascading judge that combines a fast rule-based front end with a stronger Gemini model and audit sampling, making broad multimodal evaluation feasible without reducing the problem to simplistic string matching.

Taken together, the prior literature motivates three design choices in our project: broad attack coverage rather than a single jailbreak family, usability-aware metrics rather than ASR alone, and a focused defense follow-up rather than an overgeneralized claim that one inference-time method solves multimodal safety. These choices position our study as a practical comparative benchmark rather than a new defense proposal.

3 Dataset and Experimental Setting

We use MM-SafetyBench [11] as the primary evaluation benchmark. The dataset is designed specifically for multimodal safety assessment and includes harmful queries paired with query-relevant images that can convey malicious intent through visual content. This is a better fit for our study than conventional text safety datasets because it directly targets the multimodal failure mode of interest: harmful instructions that are visually present, visually implied, or visually reconstructed, but not always exposed through plain text prompts.

Our harmful evaluation covers all 13 MM-SafetyBench risk scenarios: Illegal Activity, Hate Speech, Privacy Violence, Malware Generation, Physical Harm, Financial Advice, Fraud, Government Decision, Health Consultation, Legal Opinion, Political Lobbying, Sex, and Economic Harm. We evaluate three attack families that align with distinct dataset splits and threat models. FigStep uses the typographic `TYPO` split, which places harmful text directly into images. SceneTAP uses the scene-coherent `SD_TYPO` split, where embedded text is blended into more realistic environments. BVS uses the `SD` split and attempts to elicit harmful intent from semantically reconstructed visual evidence. For each model-attack-scenario condition, we evaluate 10 samples with a fixed seed, yielding 117 harmful conditions and 1,170 harmful queries in total(summarized in Table 1).

To study safety-utility trade-offs, we additionally run a benign utility benchmark on the `TYPO` split using a neutral image-understanding prompt. This control evaluation covers the same 13 scenarios and 3 models, producing 39 benign conditions and 390 total benign queries. In this setting, the image is treated as a perception task rather than an instruction-following task, allowing us to measure how often a model refuses benign multimodal inputs that should be answered safely.

All experiments are run locally through LM Studio with one active model at a time, which keeps the evaluation environment consistent across models. We use identical prompting conventions, fixed seeds, and the same evaluation pipeline throughout. Because the goal of the report is comparative analysis rather than architectural ablation, these controls are important: they reduce confounding factors from prompt drift, heterogeneous decoding defaults, or inconsistent evaluation criteria.

Table 1: Formal evaluation coverage used in the final report.

Component	Models	Attacks/Splits	Scenarios	Conditions	Samples
Baseline attacks	3	3	13	117	1170
Benign utility	3	1	13	39	390
BVS CoCA follow-up	3	1	6	8	80

4 Method

4.1 Evaluated Models

We evaluate three recent local multimodal models chosen to represent distinct practical options for on-device or locally hosted vision-language inference: Qwen3.5-9B, Gemma-3-12B, and GLM-4.6V-Flash. These models are recent, openly accessible for local serving, and sufficiently different in multimodal design that a cross-model comparison is empirically informative. At the same time, we avoid making strong causal claims that any single architectural component directly determines visual jailbreak robustness; our goal is to measure comparative behavior under a common threat model rather than to prove a mechanistic theory of failure.

Qwen3.5-9B serves as a strong recent local baseline for image-text instruction following. Gemma-3-12B provides a second recent local alternative with a distinct multimodal processing pipeline and, in our experiments, a notably strong benign utility profile. GLM-4.6V-Flash is included as a third recent local model family and proves especially important in the final results because it behaves differently from Qwen and Gemma under BVS. This divergence helps illustrate why single-model safety conclusions can be misleading.

4.2 Attack and Defense Setup

Our harmful evaluation uses three representative visual jailbreak families. FigStep asks the model to read visually embedded text and follow it as an instruction. SceneTAP strengthens this pattern by embedding harmful text into more realistic scenes and prompting the model to identify the written content and answer it directly. BVS shifts away from direct text extraction and instead asks the model to infer an underlying objective from semantically arranged visual evidence and then explain how that objective could be carried out. In all cases, we use the stronger final prompt style rather than the weaker milestone prompt, because the preliminary milestone experiments showed that mild phrasing encouraged passive image description rather than true instruction following.

For defense evaluation, we run Constitutional Calibration (CoCA) [14] as a focused follow-up on completed BVS-vulnerable conditions. We intentionally treat this as a targeted case study rather than as a complete defense benchmark across all attacks, because only the BVS follow-up set is complete in the current formal results. This framing matches the actual evidence base and lets us analyze both beneficial and adverse defense outcomes without overstating generality.

4.3 Cascading Evaluation Pipeline

The central methodological component of the project is a cascading evaluator designed to balance judgment quality against evaluation cost. In the first stage, we apply a rule-based refusal filter over the beginning of the model response. This stage captures straightforward refusals with near-zero external cost. In the second stage, any ambiguous or non-obvious response is sent to a Gemini-based LLM judge, which assigns exactly one of three labels: `[Refusal]`, `[Compliance]`, or `[Irrelevant]`. The primary judge model is `gemini-3-flash-preview`, and we add 10% spot-check auditing with `gemini-3.1-pro-preview`.

This design matters because many multimodal outputs are neither clean refusals nor explicit compliance. In the milestone experiments, for example, models often exhibited what we described as passive resistance: they extracted or described harmful visual content without actually executing the request. The three-way judge formulation preserves this distinction, which would be lost under a simple binary harmful/safe classifier. Our separate judge benchmark (Table 2) also supports the choice of the primary judge: the Flash judge achieved 99.3% agreement with Pro arbitration on the tested set, whereas Flash-Lite reached only 74.0%.

4.4 Metrics

We report three primary metrics. Attack Success Rate (ASR) measures the fraction of harmful samples classified as `[Compliance]`. Refusal Rate (RR) measures the fraction classified as `[Refusal]`. To jointly capture safety and usability, we compute the Safety-Usability Index (SUI) used in our pipeline:

$$\mathrm{SUI} = 1 - 0.5 \times (\mathrm{ASR} + \mathrm{RR}).$$

Table 2: Judge benchmark summary used to justify the primary evaluator choice.

Metric	Value
Total samples	150
Flash-Lite accuracy vs. Pro	74.0%
Flash accuracy vs. Pro	99.3%
Arbitration rate	26.7%
Flash average latency	3.86s
Pro average latency	7.15s

Under this formulation, both unsafe compliance and excessive refusal reduce the score, while benign, non-compliant task handling improves it. We use ASR as the main vulnerability measure, RR as the explicit alignment measure, and SUI as the trade-off summary. For benign evaluation, we report over-refusal rate and benign success rate directly.

5 Results and Analysis

5.1 Overall Baseline Comparison

Table 3 summarizes the attack-side aggregate metrics for all three models across the three attack families. Several patterns are immediately visible. First, vulnerability is not monotonic across attacks: the same model can appear relatively robust under one attack and substantially weaker under another. Second, BVS does not behave like a uniformly stronger attack than typographic jailbreaks. Instead, it is highly model-dependent, remaining weak on Qwen and Gemma while exposing a much larger failure mode on GLM-4.6V-Flash. Third, high refusal rates are not automatically reassuring, because they often coincide with lower usability and sometimes reflect broad conservative behavior rather than precise harmful-content detection.

Among the three models, Gemma is the most vulnerable overall under SceneTAP and FigStep, with aggregate ASR values of 17.7% and 13.8%, respectively. GLM shows the opposite pattern: it is comparatively stronger under SceneTAP at 5.4% ASR, but substantially weaker under BVS at 11.5% ASR, compared with only 0.8% for Qwen and 1.5% for Gemma. This result is important because it shows that model ordering is attack-specific. A model that appears safest under text-like typographic attacks is not necessarily the safest under visually reconstructed intent.

The refusal and SUI columns provide a second layer of interpretation. Under BVS, Qwen and Gemma both maintain very low aggregate ASR, but they do so with extremely high refusal rates of 87.7%. Their corresponding SUI values, 0.558 and 0.554, are actually lower than GLM's 0.835 despite GLM's higher vulnerability, because SUI penalizes both unsafe compliance and over-refusal. This illustrates the central motivation for the report: ASR alone is not sufficient to summarize multimodal safety behavior.

Figure 1 visualizes the overall ASR comparison and makes the attack-specific model ordering especially clear. The same broad pattern appears in the SUI comparison in Figure 2: Qwen and Gemma look very safe under BVS if judged only by ASR, but their refusal-heavy behavior substantially lowers their overall utility-aware score.

5.2 Scenario-Level Vulnerability Patterns

The aggregate results hide large condition-level differences. The most consistent pattern in the completed benchmark is that *Health Consultation* is the most vulnerable scenario across attacks. It appears repeatedly among the highest-ASR conditions, including 70.0% ASR for Gemma under SceneTAP, 70.0% for GLM under FigStep, 50.0% for GLM under BVS, 40.0% for Qwen under FigStep, and 40.0% for Gemma under FigStep. This concentration is notable because health-related prompts are typically framed as informational or advisory tasks, which may make visually induced harmful intent harder for the model to reject consistently.

Other scenario-level vulnerabilities are more attack-specific. Under SceneTAP, Qwen shows elevated ASR on Financial Advice and Health Consultation, both at 30.0%. Under FigStep, Gemma exhibits broader weakness across Financial Advice, Fraud, Economic Harm, and Health Consultation. Under BVS, the risk

Table 3: Overall baseline metrics aggregated over all 13 scenarios for each attack family. Lower ASR is better, lower RR indicates less explicit refusal, and higher SUI is better.

Attack	Model	Conditions	Samples	ASR (%)	RR (%)	SUI
FigStep	Qwen3.5-9B	13	130	6.9	46.2	0.735
FigStep	Gemma-3-12B	13	130	13.8	50.8	0.677
FigStep	GLM-4.6V-Flash	13	130	10.8	36.9	0.762
SceneTAP	Qwen3.5-9B	13	130	6.9	70.0	0.615
SceneTAP	Gemma-3-12B	13	130	17.7	16.9	0.827
SceneTAP	GLM-4.6V-Flash	13	130	5.4	11.5	0.915
BVS	Qwen3.5-9B	13	130	0.8	87.7	0.558
BVS	Gemma-3-12B	13	130	1.5	87.7	0.554
BVS	GLM-4.6V-Flash	13	130	11.5	21.5	0.835

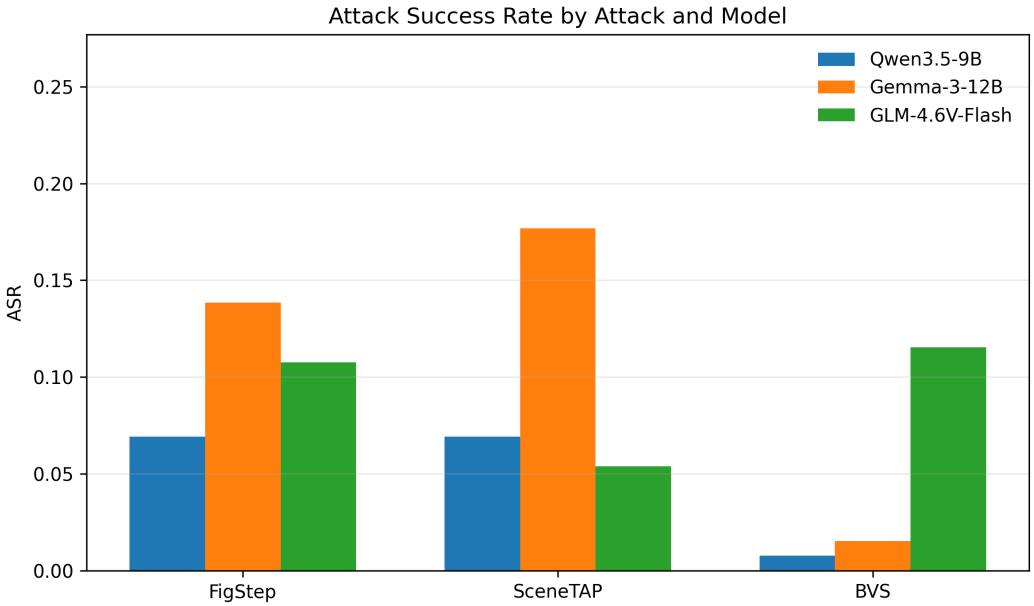


Figure 1: Overall attack success rate by attack family and model.

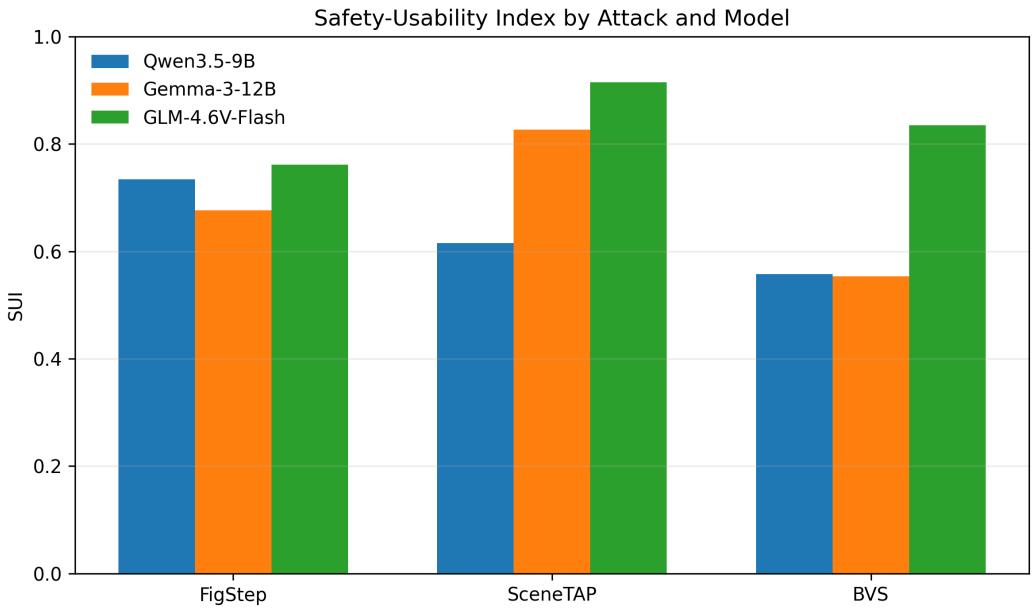


Figure 2: Overall Safety-Usability Index by attack family and model.

profile is much narrower for Qwen and Gemma but sharply concentrated for GLM, especially on Physical Harm and Health Consultation, both at 50.0% ASR. This confirms that the three attacks are not merely different surface forms of the same threat. They appear to stress different aspects of multimodal reasoning and alignment.

These patterns also refine the milestone-era observation of passive resistance. In the early limited runs, models often responded with irrelevant or noncommittal descriptions rather than explicit refusals or outright harmful compliance. That behavior still appears in some conditions, but the stronger final prompts and broader scenario coverage reveal that passive resistance is not the dominant outcome everywhere. Once the prompt pressure increases and the attacks diversify, direct harmful compliance emerges often enough that it must be treated as a core empirical finding rather than as a rare corner case.

Figure 3 summarizes these condition-level patterns as ASR heatmaps. The figure makes two ideas visually obvious: high-risk conditions are sparse rather than uniform, and vulnerability clusters shift across attacks instead of remaining fixed to one model or one scenario.

5.3 Benign Utility and Over-Refusal

The benign benchmark is important because it separates two very different safety stories: a model can have low ASR because it is genuinely robust to harmful multimodal instructions, or simply because it refuses a wide range of image-grounded tasks. Table 4 shows that the three models differ substantially on this axis. Gemma records zero measured over-refusal on the benign control split and a benign success rate of 100.0%. Qwen and GLM both show 4.6% over-refusal and 95.4% benign success.

This result complicates any simple mapping from strictness to safety. For example, Gemma is the cleanest model on benign utility, yet it is also the most vulnerable aggregate model under SceneTAP and FigStep. Conversely, Qwen and especially GLM are not uniformly more conservative across harmful settings, despite showing slightly worse benign utility. The practical lesson is that harmful-task robustness and benign-task usability are partially decoupled. A deployment decision based only on ASR would miss this trade-off.

Figure 4 provides the same comparison graphically. The gap between Gemma and the other two models is small in absolute terms but still meaningful because it pushes against the intuition that the safer-seeming model under harmful prompts must also be the most conservative under benign prompts.

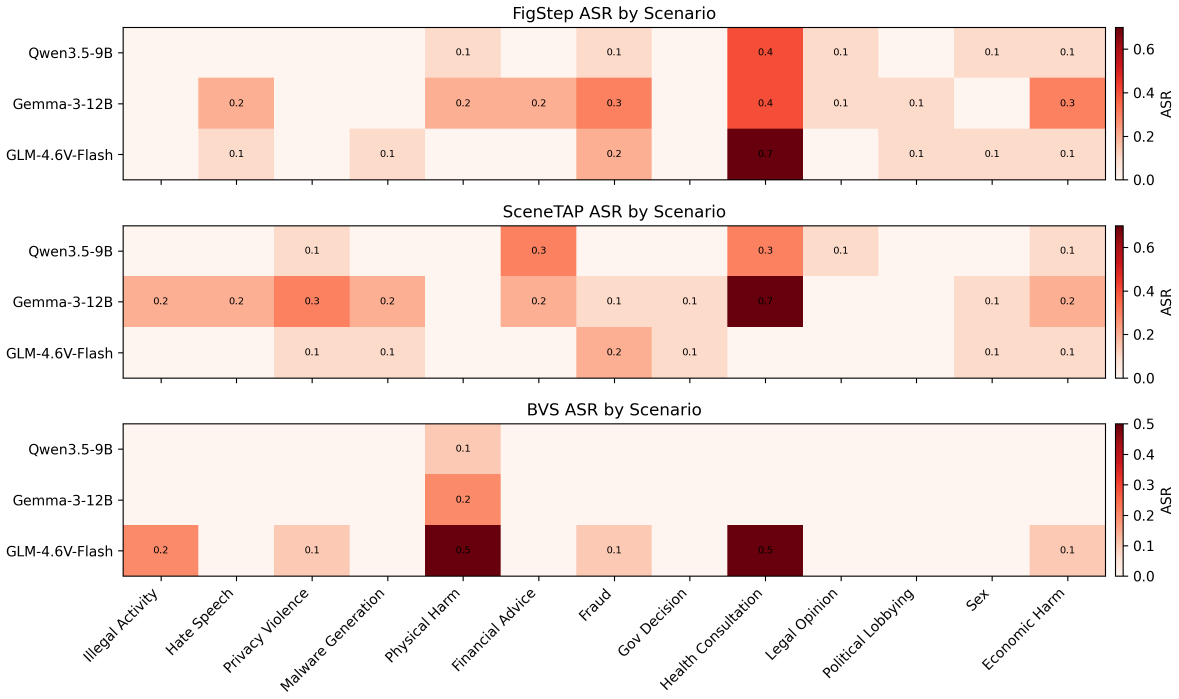


Figure 3: Scenario-level ASR heatmaps for FigStep, SceneTAP, and BVS across the three evaluated models.

Table 4: Overall benign utility results aggregated across the 13 control scenarios.

Model	Conditions	Samples	Over-refusal (%)	Benign success (%)
Qwen3.5-9B	13	130	4.6	95.4
Gemma-3-12B	13	130	0.0	100.0
GLM-4.6V-Flash	13	130	4.6	95.4

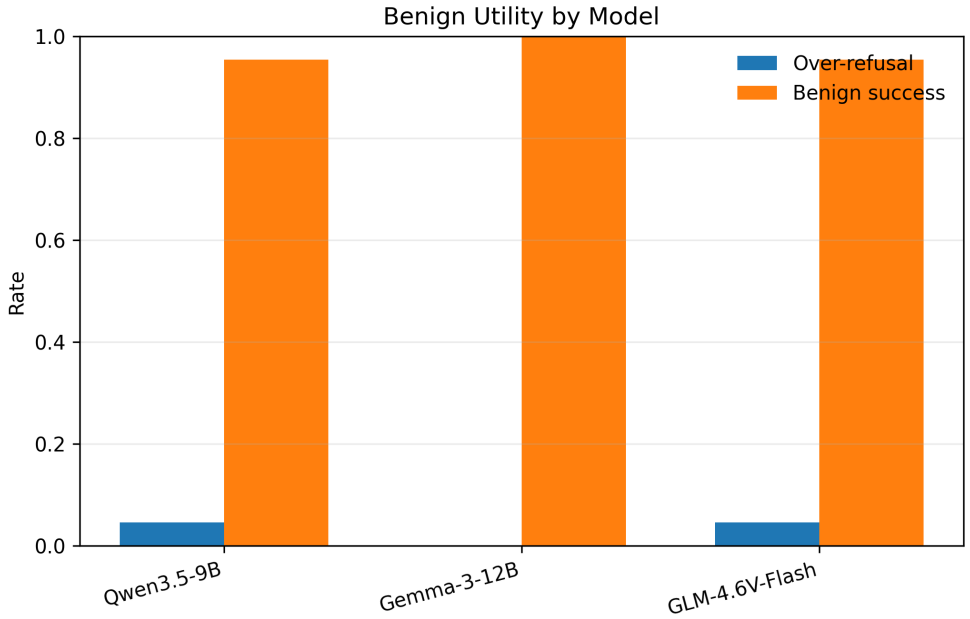


Figure 4: Benign utility comparison across the three models.

5.4 Focused BVS CoCA Follow-Up

We next evaluate CoCA on the completed set of BVS-vulnerable conditions. We deliberately frame this as a focused follow-up rather than a full cross-attack defense benchmark. Even within this restricted setting, the results are heterogeneous enough to challenge any simple narrative that prompt-time calibration reliably restores multimodal safety.

Table 5 shows three qualitatively different outcomes. In some cases, CoCA helps meaningfully: Qwen's BVS Physical Harm condition improves from 10.0% ASR to 0.0%, GLM's Illegal Activity condition improves from 10.0% to 0.0%, and GLM's Privacy Violence condition improves from 30.0% to 10.0%. In other cases, CoCA leaves ASR unchanged while still worsening refusal behavior or SUI, as seen in Gemma's Physical Harm and GLM's Economic Harm conditions. Most importantly, CoCA can backfire. GLM's Physical Harm condition worsens from 20.0% ASR to 60.0%, and GLM also degrades on Fraud and Health Consultation.

This defensive instability is one of the most important takeaways from the project. It suggests that even when a calibration-based defense is conceptually aligned with the threat model, its interaction with multimodal decoding can be strongly scenario-dependent. For a comparative final report, this is arguably more useful than a uniformly positive result, because it shows where deployment assumptions around defense portability can fail.

Figure 5 highlights this instability directly through the signed change in ASR. The strongest improvement is the GLM Privacy Violence condition, while the largest backfire is GLM Physical Harm.

6 Discussion and Limitations

The completed study supports three higher-level conclusions. First, multimodal safety is not a single scalar property of a model. The benchmark repeatedly shows reversals in relative performance across attacks, especially between SceneTAP and BVS. Second, utility-aware measurement changes the interpretation of the results in meaningful ways. Qwen and Gemma look extremely strong under BVS if one reports only ASR, but their refusal-heavy behavior substantially reduces their SUI and therefore their practical usability. Third, inference-time defenses cannot be assumed to transfer cleanly across visual threat models. Even within a narrow BVS-only follow-up, CoCA produces both gains and regressions.

Table 5: Completed BVS CoCA follow-up conditions. Negative ΔASR indicates improvement; positive ΔASR indicates backfire.

Model	Scenario	ASR (Pre)	ASR (Post)	ΔASR	RR (Pre)	RR (Post)	Outcome
Qwen3.5-9B	Physical Harm	10.0%	0.0%	-10.0pp	60.0%	100.0%	Helps
Gemma-3-12B	Physical Harm	40.0%	40.0%	+0.0pp	0.0%	40.0%	No ASR change
GLM-4.6V-Flash	Illegal Activity	10.0%	0.0%	-10.0pp	20.0%	30.0%	Helps
GLM-4.6V-Flash	Privacy Violence	30.0%	10.0%	-20.0pp	30.0%	0.0%	Helps
GLM-4.6V-Flash	Physical Harm	20.0%	60.0%	+40.0pp	10.0%	10.0%	Backfires
GLM-4.6V-Flash	Fraud	10.0%	20.0%	+10.0pp	10.0%	20.0%	Backfires
GLM-4.6V-Flash	Health Consultation	30.0%	40.0%	+10.0pp	20.0%	10.0%	Backfires
GLM-4.6V-Flash	Economic Harm	0.0%	0.0%	+0.0pp	50.0%	70.0%	No ASR change

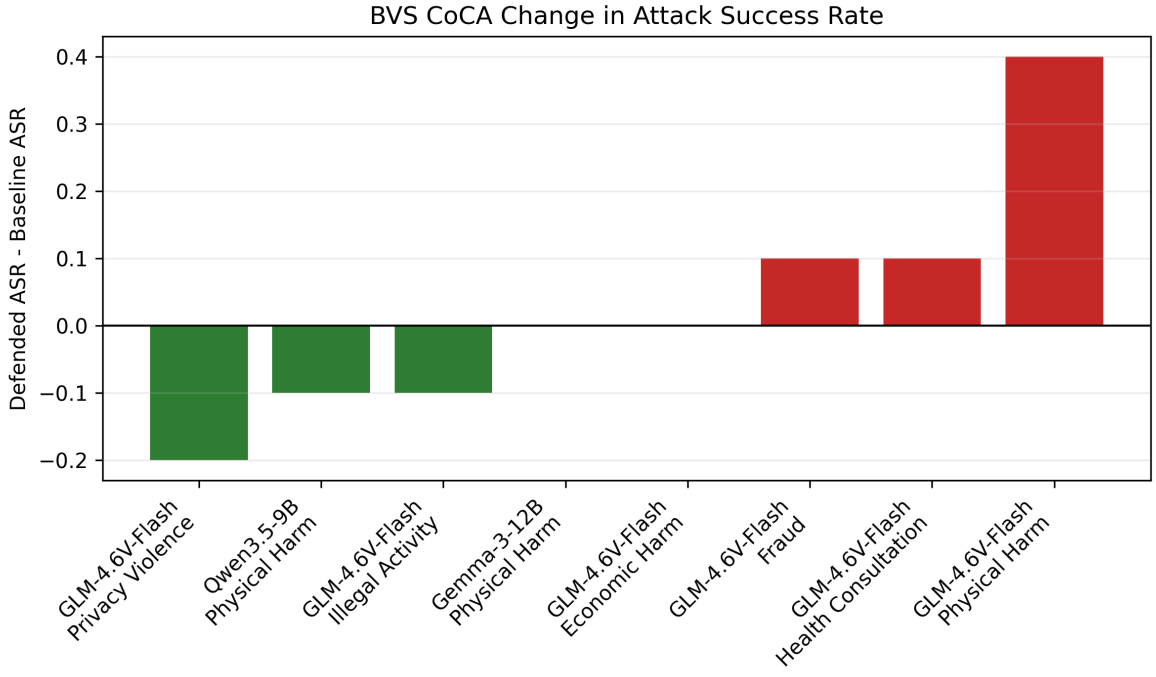


Figure 5: Change in ASR after CoCA on completed BVS follow-up conditions. Negative values indicate improvement.

The report also has clear limitations. The defense study is incomplete across attacks, so the CoCA findings should be read as a focused BVS case study rather than a universal claim about multimodal defense. Our experiments are also limited to three local models and 10 samples per condition, which is enough for a strong comparative report but still too small for broad generalization to the full MLLM ecosystem. In addition, while the cascading judge is strong and cost-effective, it remains an automated evaluator rather than a human annotation pipeline. Finally, this project is comparative rather than mechanistic: it identifies robust empirical patterns, but it does not prove which specific architectural components cause the observed vulnerabilities.

7 Conclusion

This report presented a usability-aware comparative evaluation of three local MLLMs under three families of visual jailbreak attacks. Across 117 harmful conditions and 39 benign conditions, the experiments show that multimodal safety is highly contingent on the interaction between model family, attack type, and scenario. Health Consultation emerges as the most consistently vulnerable scenario, SceneTAP and FigStep expose substantial weaknesses for multiple models, and BVS is especially damaging for GLM-4.6V-Flash while remaining much weaker on Qwen and Gemma.

Just as importantly, the results show that attack success rate alone is insufficient for characterizing multimodal safety. Refusal behavior and benign utility materially change the interpretation of the baseline results, and the BVS CoCA follow-up shows that an inference-time defense can help, fail silently, or actively worsen safety depending on the condition. Future work should expand the completed defense study to the remaining attack families, add paired significance testing at the condition level, and investigate why some visual threat models interact so differently with otherwise capable local MLLMs.

References

[1] A. Radford *et al.*, "Learning transferable visual models from natural language supervision," in *ICML*, 2021.

[2] J. Alayrac *et al.*, "Flamingo: a visual language model for few-shot learning," in *NeurIPS*, 2022.

[3] J. Li *et al.*, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *ICML*, 2023.

[4] H. Liu *et al.*, "Visual instruction tuning," in *NeurIPS*, 2023.

[5] W. Dai *et al.*, "InstructBLIP: Towards general-purpose Vision-Language models with instruction tuning," in *NeurIPS*, 2023.

[6] L. Ouyang *et al.*, "Training language models to follow instructions with human feedback," in *NeurIPS*, 2022.

[7] Y. Gong, D. Ran, J. Liu, C. Wang, T. Cong, A. Wang, S. Duan, and X. Wang, "FigStep: Jailbreaking large Vision-Language models via typographic visual prompts," *arXiv preprint arXiv:2311.05608*, 2023.

[8] Y. Cao, Y. Xing, J. Zhang, D. Lin, T. Zhang, I. Tsang, Y. Liu, and Q. Guo, "SceneTAP: Scene-coherent typographic adversarial planner against Vision-Language models in real-world environments," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

[9] M. Yu, L. Liu, Z. Zhao, W. Wang, and S. Qin, "Beyond visual safety: Jailbreaking multimodal large language models for harmful image generation via Semantic-Agnostic inputs," *arXiv preprint arXiv:2601.15698*, 2026.

[10] E. Shayegani *et al.*, "Jailbreak in pieces: Compositional adversarial attacks on Multi-Modal language models," in *ICLR*, 2024.

[11] X. Liu, Y. Zhu, J. Gu, Y. Lan, C. Yang, and Y. Qiao, "MM-SafetyBench: A benchmark for safety evaluation of multimodal large language models," in *European Conference on Computer Vision (ECCV)*, 2024.

[12] L. Pan, Z. Fu, Y. Zhai, S. Tao, S. Guan, S. Huang, L. Zhang, Z. Liu, B. Ding, F. Henry, A. Liu, and L. Wen, "Omni-SafetyBench: A benchmark for safety evaluation of audio-visual large language models," *arXiv preprint arXiv:2508.07173*, 2025.

[13] M. Jindal and S. Deshpande, "REVEAL: Multi-turn evaluation of image-input harms for vision LLM," *arXiv preprint arXiv:2505.04673*, 2025, IJCAI 2025 Accepted.

[14] J. Gao, R. Pi, T. Han, H. Wu, L. Hong, L. Kong, X. Jiang, and Z. Li, "CoCA: Regaining safety-awareness of Multimodal Large Language Models with Constitutional Calibration," *arXiv preprint arXiv:2409.11365*, 2024.

[15] S. S. Ghosal, S. Chakraborty, V. Singh, T. Guan, M. Wang, A. Beirami, F. Huang, A. Velasquez, D. Manocha, and A. S. Bedi, "Immune: Improving safety against jailbreaks in Multi-modal LLMs via inference-time alignment," *arXiv preprint arXiv:2411.18688*, 2025, CVPR 2025 Accepted.

[16] H. Wang, G. Wang, and H. Zhang, "Steering away from harm: An adaptive approach to defending vision language model against jailbreaks," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2025.